

# VAIBHAV SETHI

Data Analyst · Applied Statistics · Machine Learning · Risk Analytics

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## SUMMARY

UBC Statistics & Physics student (coursework complete Dec 2026) who builds analytics products end to end: SQL feature pipelines on 19M+ records, validated and calibrated machine-learning models, and interactive dashboards that non-technical stakeholders can use. Flagship project stress-tests a \$35B portfolio of 350K+ real Freddie Mac mortgages under the Federal Reserve's 2026 scenarios, shipped as a live web dashboard with full model documentation. Strong across SQL, Python, R, Excel, and Power BI. Open to Fall 2026 co-op; available full-time from January 2027.

## TECHNICAL SKILLS

**Data & Programming** SQL (window functions, CTEs — DuckDB) · Python (pandas, NumPy, scikit-learn, XGBoost, statsmodels) · R (tidyverse, ggplot2, Shiny) · SAS (PROC LOGISTIC, PROC SQL) · Excel (financial modeling)

**Visualization & BI** Power BI (star schema) · Tableau · Shiny · Interactive web dashboards (HTML/JS)

**Statistics & ML** Regression & GLMs · Time Series · Survival / Hazard Models · PCA & Clustering · Probability Calibration · Imbalanced Classification · Model Validation (ROC / PR-AUC, Gini, KS) · SHAP Explainability

**Risk & Workflow** Expected Loss ( $PD \times LGD \times EAD$ ) · Stress Testing (CCAR / DFAST-style) · Model Cards & Documentation · Git & GitHub · Jupyter · Parquet

## PROJECTS

**Mortgage Stress-Testing Engine** — Python · SQL · R · SAS · DuckDB · XGBoost · Power BI 2026  
Live dashboard, case study & model card: [vaibhavsethi01.github.io/portfolio](https://vaibhavsethi01.github.io/portfolio)

- Engineered features across 19M loan-month records in DuckDB from 350K+ real Freddie Mac mortgages; trained a probability-of-default model validated out-of-time by vintage (Gini 0.66 · KS 0.50) and calibrated it, cutting Brier score 63% (0.175 → 0.064).
- Projected losses over a 13-quarter horizon under the Fed's 2026 baseline vs. severely-adverse scenarios — \$65M vs. \$352M expected loss on a \$35B book (5.5× stress multiple) — and shipped a live, in-browser dashboard and stress simulator.
- Documented data, evaluation, and limitations in an SR 11-7-style model card; reproduced the model in SAS and R and delivered a Power BI star-schema export.

**Banking Risk Platform — Credit Scoring & Expected Loss** — R · XGBoost · Shiny · SHAP 2024 – 2025  
Code: [github.com/vaibhavsethi01/banking-risk-platform](https://github.com/vaibhavsethi01/banking-risk-platform)

- Built an end-to-end credit-scoring application in R/Shiny: XGBoost benchmarked against logistic regression on heavily imbalanced data, evaluated on PR-AUC, and Platt-calibrated so scores behave as true probabilities.
- Simulated portfolio expected loss ( $PD \times LGD \times EAD$ ) and served per-prediction SHAP reason codes, with single and batch CSV scoring built for non-technical users.

**LIGO Gravitational-Wave Glitch Classifier** — Python · scikit-learn · Random Forest 2026  
Classified 7,966 detector-noise samples across 22 classes with 67:1 imbalance; tuned a Random Forest via grid search to 88% accuracy and 0.81 macro-F1, with cross-validation and test scores within 0.01.

**Vancouver Housing Price Prediction** — R · Python · ggplot2 · Time Series 2023  
Modeled residential prices from 10+ years of transactions joined with macro indicators (rates, CPI, income); engineered spatial and temporal features, validated on held-out sets via RMSE / MAE, and communicated drivers to a non-technical audience.

## EXPERIENCE

**Printing Services Representative**, Staples Canada — Vancouver, BC Dec 2022 – Jun 2023  
Delivered 50–70 client print orders per shift against hard deadlines; translated customer requirements into production specs and served as first point of resolution for equipment failures.

## EDUCATION

**University of British Columbia** — B.Sc., Combined Major in Statistics and Physics **Coursework complete Dec 2026 · Convocation May 2027**  
Selected coursework: Regression & GLMs, Time Series, Machine Learning, Statistical Inference, Multivariate Analysis, Experimental Design, Statistical Computing (R & Python)

## CERTIFICATIONS & ACTIVITIES

Bloomberg Market Concepts (BMC) · UBC Data Science Club — Member (2025 – Present)